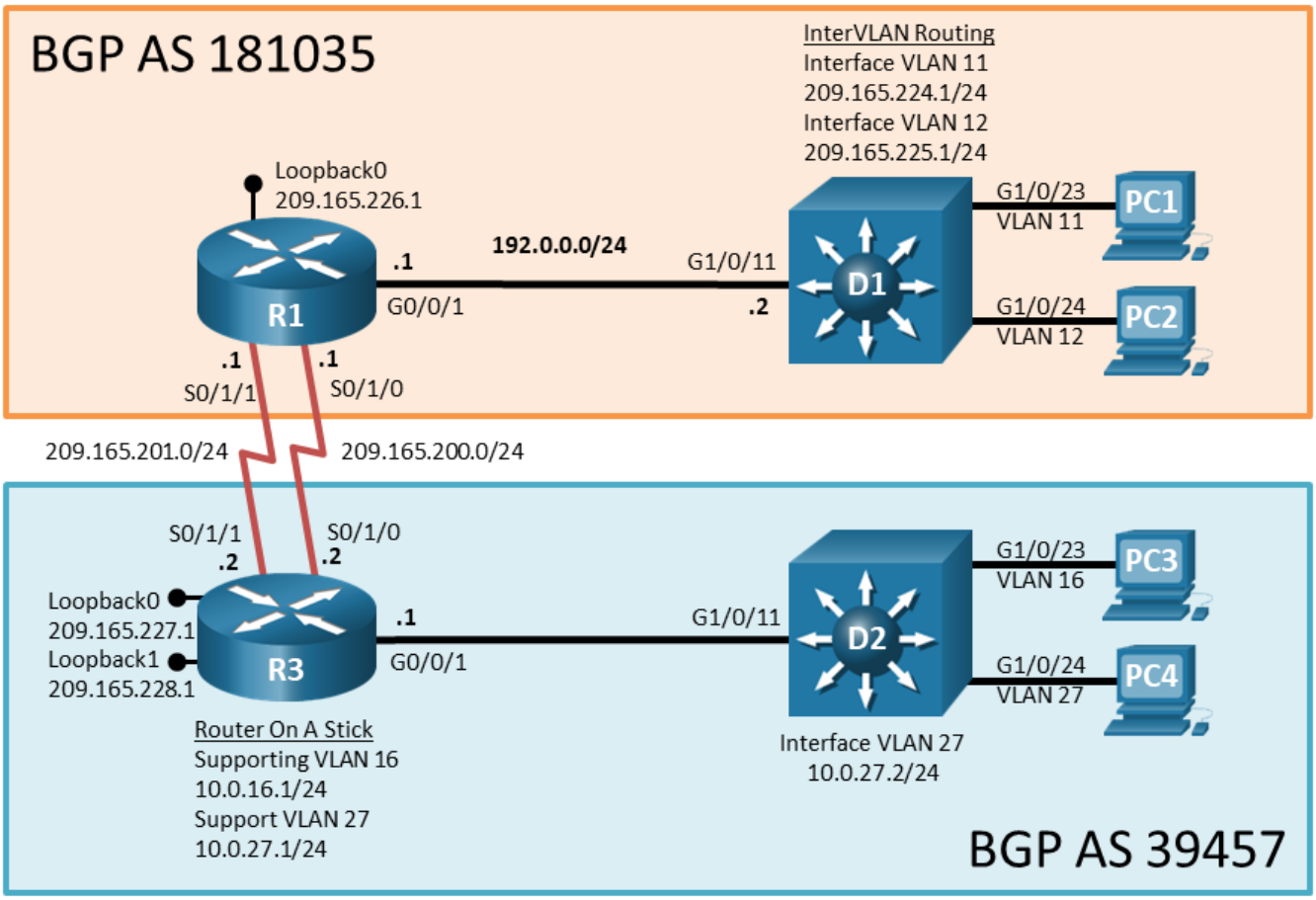
Lab - Troubleshoot IPv4 ACLs

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask
R1	G0/0/1	192.0.0.1	255.255.255.0
	S0/1/0	209.165.200.1	255.255.255.0
	S0/1/1	209.165.201.1	255.255.255.0
	Loopback0	209.165.226.1	255.255.255.0
R3	G0/0/1.16	10.0.16.1	255.255.255.0
	G0/0/1.27	10.0.27.1	255.255.255.0

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Device	Interface	IP Address	Subnet Mask
	S0/1/0	209.165.200.2	255.255.255.0
	S0/1/1	209.165.201.2	255.255.255.0
	Loopback0	209.165.227.1	255.255.255.0
	Loopback1	209.165.228.1	255.255.255.0
D1	G1/0/11	192.0.0.2	255.255.255.0
	VLAN 11	209.165.224.1	255.255.255.0
	VLAN 12	209.165.225.1	255.255.255.0
PC1	NIC	DHCP	
PC2	NIC	DHCP	
PC3	NIC	DHCP	
PC4	NIC	DHCP	

Objectives

Troubleshoot network issues related to the configuration and operation of ACLs for IPv4.

Background / Scenario

In this topology, R1 and D1 are OSPF neighbors, while R1 and R3 are BGP neighbors. Switch D1 provides inter-VLAN routing for two subnets. R3 provides inter-VLAN routing for two subnets, and switch D2 provides connectivity for the two VLANs supporting those subnets. The BGP relationship between R1 and R3 is established using EBGP multihop between the router's respective Loopback 0 interfaces. You will be loading configurations with intentional errors onto the network. Your tasks are to FIND the error(s), document your findings and the command(s) or method(s) used to fix them, FIX the issue(s) presented here and then test the network to ensure both of the following conditions are met:

1) the complaint received in the ticket is resolved
2) full reachability is restored

Note: The routers used with CCNP hands-on labs are Cisco 8200L with Cisco IOS XE Release 17.12.05c (universalk9 image). The switches used in the labs are Cisco Catalyst 9300L with Cisco IOS XE Release 17.12.05 (universalk9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: Make sure that the devices have been erased and have no startup configurations. If you are unsure, contact your instructor.

Required Resources

- 2 Routers (Cisco 8200L with Cisco IOS XE Release 17.12.05c universal image or comparable)
- 2 Switches (Cisco 9300L with Cisco IOS XE Release 17.12.05 universal image or comparable)
- 4 PCs (Choice of operating system with terminal emulation program installed)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet and serial cables as shown in the topology

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Instructions

Part 1: Trouble Ticket 32.1.1.1

Scenario:

A Sunday night change window was used by an external security consultant to bring R1 and R3 into alignment with RFC 1918 policies. The work included updates to ACLs and NAT policy on the edge path between the campus and the WAN. By 06:45 Monday morning, Operations reports that a business-critical workflow between PC1 in VLAN 11 and PC3 in VLAN 27 is failing. Pings time out and the application no longer connects. OSPF adjacencies look stable and the BGP session between R1 and R3 is established, but users are blocked from processing orders. Your task is to investigate and restore the PC1-PC3 communication while preserving the intended security posture from the change.

Be sure you are using devices that have been connected and prepared by your instructor. Use the commands listed below to load the configuration files for this trouble ticket:

Device	Command
R1	`copy flash:/enarsi/32.1.1.1-r1-config.txt running-config`
R3	`copy flash:/enarsi/32.1.1.1-r3-config.txt running-config`
D1	`copy flash:/enarsi/32.1.1.1-d1-config.txt running-config`
D2	`copy flash:/enarsi/32.1.1.1-d2-config.txt running-config`

- PCs 1, 2, 3, and 4 receive their addressing via DHCP for IPv4.
- PC3 should be able to ping PC1. However, PC1 should not be able to ping PC3.
- Passwords on all devices are **cisco12345**. If a username is required, use **admin**.
- When you have fixed the ticket, change the MOTD on EACH DEVICE using the following command:

  banner motd # This is $(hostname) FIXED from ticket <ticket number> #
- Then save the configuration by issuing the **wri** command (on each device).
- Inform your instructor that you are ready for the next ticket.
- After the instructor approves your solution for this ticket, issue the **reset.now** privileged EXEC command. This script will clear your configurations and reload the devices.

Part 2: Trouble Ticket 32.1.1.2

Scenario:

Late Friday evening, a junior network administrator “tightened” ACLs to reduce unnecessary exposure and lateral movement. Early Monday, Finance users on VLAN 12 (PC2 via DHCP) report they cannot reach two internal services at 209.165.227.1 and 209.165.228.1, which host reporting and licensing endpoints on R3. Users on VLAN 11 can still reach these addresses, suggesting a policy impact specific to VLAN 12. You need to restore PC2’s reachability to those loopback addresses without undermining the hardening intent.

Use the commands listed below to load the configuration files for this trouble ticket:

Device	Command
R1	`copy flash:/enarsi/32.1.1.2-r1-config.txt running-config`
R3	`copy flash:/enarsi/32.1.1.2-r3-config.txt running-config`

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Device	Command
D1	`copy flash:/enarsi/32.1.1.2-d1-config.txt running-config`
D2	`copy flash:/enarsi/32.1.1.2-d2-config.txt running-config`

- PCs 1, 2, 3, and 4 receive their addressing via DHCP for IPv4.
- PC2 should be able to ping 209.165.226.1 and 209.165.227.1.
- Passwords on all devices are **cisco12345**. If a username is required, use **admin**.
- When you have fixed the ticket, change the MOTD on EACH DEVICE using the following command:

  banner motd # This is $(hostname) FIXED from ticket <ticket number> #
- Then save the configuration by issuing the **wri** command (on each device).
- Inform your instructor that you are ready for the next ticket.
- After the instructor approves your solution for this ticket, issue the **reset.now** privileged EXEC command. This script will clear your configurations and reload the devices.

Part 3: Trouble Ticket 32.1.1.3

Scenario:

Over the weekend, a junior administrator attempted to enforce time-based restrictions on remote access as part of security improvements related to the external BGP peer in AS 181035. At 08:00 Monday, NOC staff report they cannot Telnet to R1 or D1 to perform routine checks and incident response, impacting morning operations. Console access is limited across sites, so daytime remote management must be restored quickly while keeping the intent of off-hours restrictions. Your job is to identify the access control misconfiguration and re-establish appropriate remote access during business hours.

Use the commands listed below to load the configuration files for this trouble ticket:

Device	Command
R1	`copy flash:/enarsi/32.1.1.3-r1-config.txt running-config`
R3	`copy flash:/enarsi/32.1.1.3-r3-config.txt running-config`
D1	`copy flash:/enarsi/32.1.1.3-d1-config.txt running-config`
D2	`copy flash:/enarsi/32.1.1.3-d2-config.txt running-config`

- PCs 1, 2, 3, and 4 receive their addressing via DHCP for IPv4.
- PC1 should be able to Telnet into R1 and D1.
- Passwords on all devices are **cisco12345**. If a username is required, use **admin**.
- When you have fixed the ticket, change the MOTD on EACH DEVICE using the following command:

  banner motd # This is $(hostname) FIXED from ticket <ticket number> #
- Then save the configuration by issuing the **wri** command (on each device).
- Inform your instructor that you are ready for the next ticket.
- After the instructor approves your solution for this ticket, issue the **reset.now** privileged EXEC command. This script will clear your configurations and reload the devices.

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